PRISMA 2020 Eligibility Stage: Quality Criteria Scoring — Section 4.2 (Feature Drift and Multi-Signal / Dual-Phenomenon Frameworks)

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1. Scope of This Scoring Pass

This report extends the Step 4 eligibility scoring to the full candidate set retrieved for Section 4.2 (Feature Drift and Multi-Signal / Dual-Phenomenon Frameworks) of the review, comprising all 11 full-text articles retained for this subsection following Step 3 screening. Each article was scored honestly against its full text rather than against title and abstract alone.

QC1 (problem statement and scope clarity) was scored against fit to this review's specific research question, namely concept drift, spatial domain shift, and continual or federated adaptation in machine learning-based IoT intrusion detection, and, within Section 4.2 specifically, against whether an article's problem statement concerns feature drift or a multi-signal or dual-phenomenon distinction relevant to the dataset-shift taxonomy of Table V. An article can be rigorous and well-scoped for a different question, such as domain-agnostic drift-detector benchmarking, wireless-network fingerprinting, or advanced-persistent-threat transfer learning, and still receive a Partial rather than full QC1 score here, because the object of appraisal is not IoT intrusion detection specifically. A Partial score of 0.5, rather than zero, was assigned wherever an article's core mechanism was genuinely transferable and substantively discussed by the manuscript, but the evaluated application domain was general anomaly detection, wireless fingerprinting or link-anomaly detection, or advanced-persistent-threat and transfer-learning detection rather than IoT-IDS network traffic.

QC4 (empirical extraction detail) was scored as 0 wherever an article's full text contained no drift-type, distributional-component, or adaptation-mechanism data extractable into this review's own extraction fields, regardless of how much extraction detail the article offered for its own, different research question. This reflects a consistent, review-specific application of the criterion rather than a per-article adjustment.

QC2 (methodological transparency) and QC3 (venue indexing and citability) were scored on their ordinary, article-intrinsic basis, since these criteria measure the rigor and citability of the work itself rather than its fit to this particular review.

2. Quality Assessment Criteria Applied

Table 1 reports the four quality criteria (QC1 to QC4) applied in this eligibility round, together with the scoring scheme used for each. A quality score is computed as the sum of applicable criterion scores divided by the number of applicable criteria, scaled to a 0 to 4 range.

Table 1. Quality assessment criteria applied to Section 4.2 candidate articles

Code	Criterion	Scoring
QC1	Problem statement and scope clarity	Yes (1) / Partial (0.5) / No (0)
QC2	Methodological transparency	Yes (1) / Partial (0.5) / No (0)
QC3	Venue indexing and citability	Yes (1) / Partial (0.5) / No (0)
QC4	Empirical extraction detail (where applicable)	Yes (1) / Partial (0.5) / No (0) / N/A

Articles published in 2020 or later were retained where their total quality score met or exceeded 2.5 out of 4. Articles published before 2020 were retained where their total quality score met or exceeded 1.5 out of 4.

3. Results

Of the 11 full-text articles assessed for Section 4.2 against the quality criteria in Section 2, 7 articles met or exceeded their applicable threshold and were retained, while 4 articles fell below their applicable threshold and were excluded. Table 2 lists each assessed article together with its per-criterion and total quality scores, applicable threshold, and eligibility decision, sorted with included articles listed first, followed by excluded articles.

Table 2. Quality criteria scoring results for Section 4.2 candidate articles, sorted by decision

No.	Citation	Year	QC1	QC2	QC3	QC4	Total	Threshold	Decision
1	Feature Drift Aware for Intrusion Detection System using Developed Variable Length Particle Swarm Optimization in Data Stream (2023)	2023	1.0	1.0	1.0	1.0	**4.0**	≥2.5	**Include**
2	Intrusion Detection in the IoT under Data and Concept Drifts: Online Deep Learning Approach (2022)	2022	1.0	1.0	1.0	1.0	**4.0**	≥2.5	**Include**
3	Unsupervised Incremental Learning with Dual Concept Drift Detection for Identifying Anomalous Sequences (2024)	2024	0.5	1.0	1.0	1.0	**3.5**	≥2.5	**Include**
4	A Representation Learning Approach to Feature Drift Detection in Wireless Networks (2026)	2026	0.5	1.0	1.0	1.0	**3.5**	≥2.5	**Include**
5	Addressing Concept Drift in IoT Anomaly Detection: Drift Detection, Interpretation, and Adaptation (2024)	2024	1.0	1.0	1.0	1.0	**4.0**	≥2.5	**Include**
6	Reinforcement Learning-Based Voting for Feature Drift-Aware Intrusion Detection: An Incremental Learning Framework (2025)	2025	1.0	1.0	1.0	1.0	**4.0**	≥2.5	**Include**

No.	Citation	Year	QC1	QC2	QC3	QC4	Total	Threshold	Decision
7	From one attack domain to another: Contrastive transfer learning with siamese networks for APT detection (2026)	2026	0.5	1.0	1.0	1.0	**3.5**	≥2.5	**Include**
8	Evolving cybersecurity frontiers: A comprehensive survey on concept drift and feature dynamics aware machine and deep learning in intrusion detection systems (2024)	2024	0.5	0.5	1.0	0.0	**2.0**	≥2.5	**Exclude**
9	DyP-CNX: A Dynamic Preprocessing-Enhanced Hybrid Model for Network Intrusion Detection (2025)	2025	0.0	1.0	1.0	0.0	**2.0**	≥2.5	**Exclude**
10	A Framework for Drift Detection and Adaptation in AI-driven Anomaly and Threat Detection Systems (2025)	2025	0.5	0.5	1.0	0.0	**2.0**	≥2.5	**Exclude**
11	A Novel Method for Drift Detection in Streaming Data Based on Measurement of Changes in Feature Ranks (2025)	2025	0.0	1.0	1.0	0.0	**2.0**	≥2.5	**Exclude**

This retained set of 7 articles constitutes the eligible pool for Section 4.2 (Feature Drift and Multi-Signal / Dual-Phenomenon Frameworks) and forms part of the input to the synthesis and thematic analysis stage.

4. Data Integrity Notes

Each article was scored against the four quality criteria on the basis of its full text, applying the review-specific interpretation of QC1 and QC4 set out in Section 1 above. The retain or exclude decision was applied strictly according to the threshold in effect for that article's publication year, namely 2.5 out of 4 for articles published 2020 or later, or 1.5 out of 4 for articles published before 2020. No article was retained or excluded on any basis other than its quality score relative to its applicable threshold.

5. Document Preparation

This report has been prepared in accordance with the quality assurance procedure prior to proceeding to the synthesis stage.

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